

FORCE

FALL · OPTICAL RATE · CARDIAC EMITTER

Technical Documentation – Smart Assistive Crutch System for Elderly Care

INSTITUTION

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ABSTRACT

FORCE is an embedded elderly-care system mounted on a standard forearm crutch. It continuously monitors heart rate, blood oxygen saturation, and body temperature via optical sensors, detects falls through a multi-stage IMU-based algorithm, and transmits live telemetry to a caregiver dashboard via BLE and a REST API server. All firmware, sensor housings, and mounting hardware not provided at the start of the contest were custom designed and 3D-printed by the team.

01 SYSTEM OVERVIEW

FORCE is a full-stack assistive health monitoring system built around a standard forearm crutch. The system spans three layers – embedded firmware on an ESP32 microcontroller, a local REST API server, and two mobile applications – forming a closed telemetry loop from sensor to caregiver screen.

SYSTEM ARCHITECTURE AT A GLANCE



DESIGN PRINCIPLES

The firmware is written in C++ using the Arduino/ESP-IDF framework and structured as a **cooperative task scheduler** – a lightweight array of `Task` structs, each carrying an interval and a function pointer, ticked by `runTaskScheduler()` in the main loop. Tasks range from 100 Hz IMU reads to 3-second temperature polls, with no RTOS overhead.

All fall detection logic executes **entirely on the ESP32**, with no cloud round-trip in the critical path. A fall alert reaches the caregiver's phone within approximately 500 ms of sensor data confirming the event – irrespective of server latency.

CUSTOM HARDWARE

Every physical component not provided at contest start was custom-designed and **3D-printed** by the team: the tip enclosure housing the IMU, ESP32, and buzzer; the handle housing integrating the MAX30102 optical window, MAX30205 probe, and capacitive touch pad, the OLED display bracket, and all PCB mounts and cable guides. The crutch frame itself is a standard forearm crutch.

ESP32-WROOM-32	MPU-6050 IMU	MAX30102 Oximeter	MAX30205 Temp	SSD1306 OLED	Piezo Buzzer
Capacitive Touch	3D-Printed Housings				

02 HARDWARE ARCHITECTURE

All sensors communicate over a shared **PC bus** (SDA = GPIO 21, SCL = GPIO 22, 100 kHz) with the exception of the touch input, which uses the ESP32's native capacitive-touch peripheral, and the buzzer, which is driven directly from a GPIO via the `tone()` API. The MPU-6050 additionally uses a dedicated interrupt line (GPIO 4) to signal data-ready at 100 Hz, eliminating polling overhead.

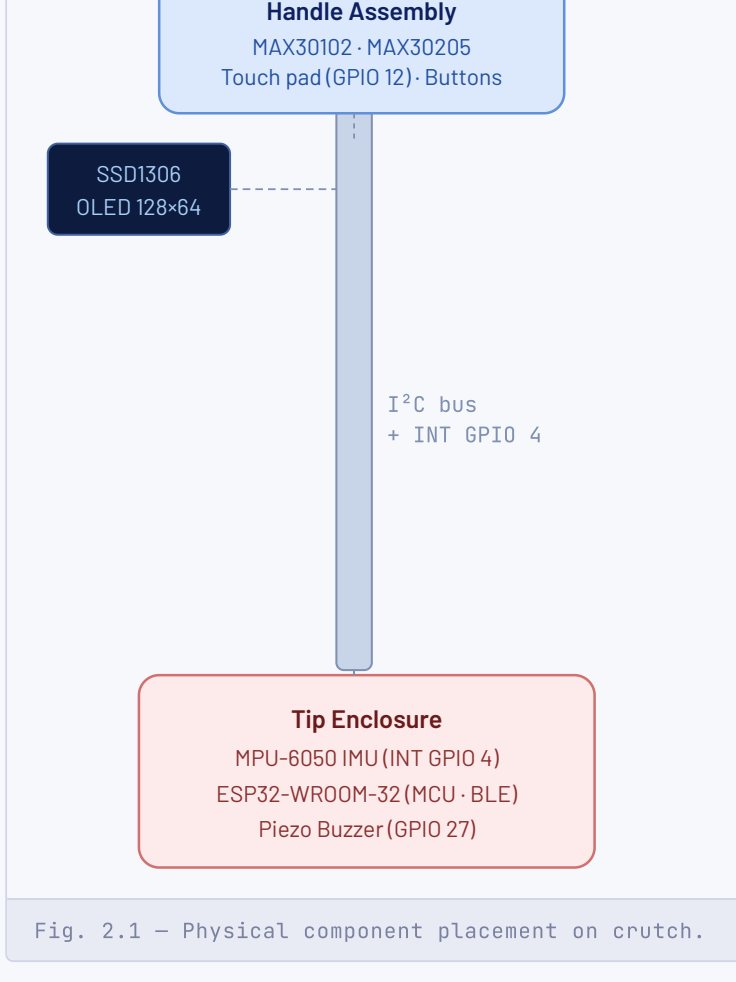
PHYSICAL PLACEMENT

HANDLE ASSEMBLY (TOP)

The MAX30102 pulse oximeter, MAX30205 body temperature sensor, and capacitive touch pad are co-located in the custom 3D-printed handle. Sustained palm contact gives the optical sensor a stable skin window and keeps the thermistor in contact with the user's hand. The touch pad confirms grip, feeding Gate 0. The panic button, OLED display, and mirror button are also handle-mounted for single-hand operation.

TIP ENCLOSURE (BOTTOM)

The MPU-6050 IMU, ESP32 microcontroller, and piezo buzzer are all co-located in the 3D-printed tip enclosure. Placing the ESP32 at the tip keeps all high-power-draw electronics away from the biometric sensors, reduces wiring length to the IMU, and maximises the IMU lever-arm – small angular displacements at the tip produce large gyroscope readings, improving fall sensitivity.



PIN ASSIGNMENT SUMMARY

GPIO	Component	Interface	Purpose
21 / 22	All PC devices	I²C SDA/SCL	Shared sensor bus (100 kHz)
4	MPU-6050	Digital interrupt	Data-ready IRQ at 100 Hz
13 / 14	SparkFun Bio Hub	Digital I/O	RESET and MFIO control lines
27	Piezo buzzer	PWM / tone()	Audible alarm output
33	Panic button	Analog input	Voltage-divider multi-button
12	Capacitive pad	ESP32 touch	Handle-grip detection
16 / 17	Status LEDs	Digital output	Gate 0 armed · BLE connected

MULTI-BUTTON ENCODING

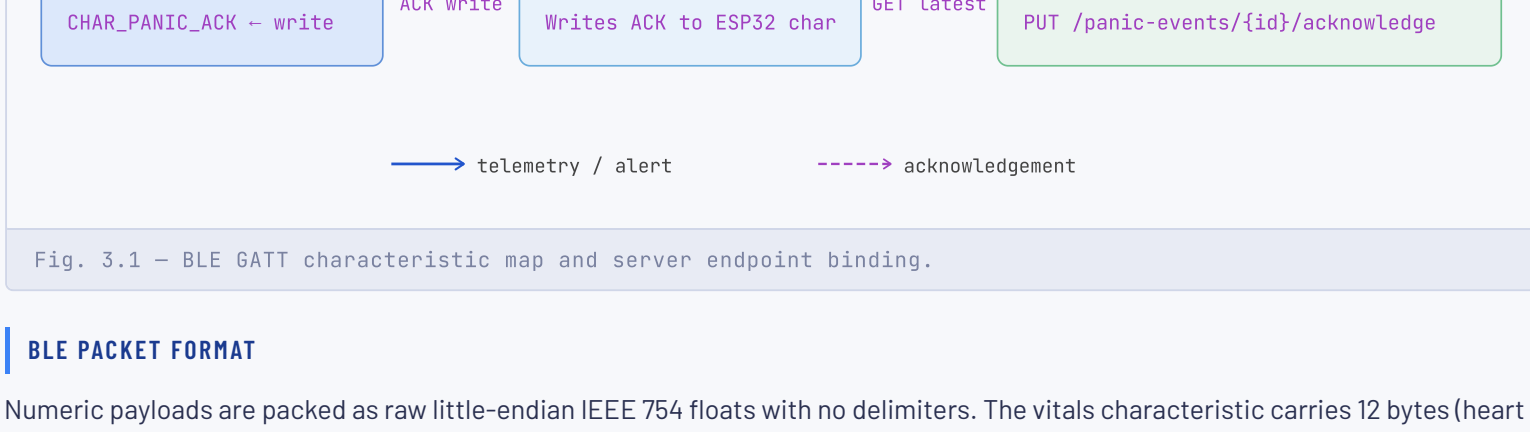
The panic button and mirror button share a single analog pin (GPIO 33) via a resistor voltage divider. An ADC reading above 3000 counts identifies the panic button (~3.3 V rail); a reading between 1500 and 3000 counts identifies the secondary button (~1.6 V). This saves a GPIO while maintaining two distinct inputs.

03 DATA FLOW ARCHITECTURE

The FORCE telemetry pipeline is a **three-hop chain**: ESP32 → Phone App → REST Server → Caregiver App. Each hop uses a different protocol optimised for its context – BLE for the low-power wireless link to the crutch, HTTP/REST for reliable server persistence, and polling REST for the caregiver dashboard.

HOP 1 – ESP32 TO PHONE APP VIA BLE

The ESP32 acts as a **BLE peripheral**, advertising the FORCE service UUID and exposing five GATT characteristics. The phone app acts as the central/client. All notify characteristics push data at 100 ms intervals; the phone writes to the acknowledgement characteristic to confirm panic receipt.



BLE PACKET FORMAT

Numeric payloads are packed as raw little-endian IEEE 754 floats with no delimiters. The vitals characteristic carries 12 bytes (heart rate, SpO₂ confidence); the motion characteristic carries 24 bytes (ax, ay, az, gx, gy, gz) with accel values pre-scaled to g and gyro values pre-scaled to °/s before packing. This minimises BLE payload size while preserving full float precision.

HOP 2 – PHONE APP TO REST SERVER

Every BLE notification received by the phone app triggers an **HTTP POST** to the corresponding REST endpoint at `372.28.189.35:8080`, authenticated with a static Bearer token. The server persists all records to a **PostgreSQL** database with timestamps. Panic events carry an `eventType` of either `PANIC_BUTTON` or `FALL_DETECTION`, written by the phone app based on the value of the BLE panic characteristic (0 = manual, 1 = fall).

HOP 3 – REST SERVER TO CAREGIVER APP

The caregiver app polls the server via HTTP GET on `/api/vital-signs/latest`, `/api/temperature/latest`, and `/api/panic-events/latest` to populate the live dashboard. When a panic event appears with `status: ACTIVE`, the caregiver app issues a `PUT /api/panic-events/{id}/acknowledge` to mark it resolved, setting `acknowledgedAt`, and transitioning status to `ACKNOWLEDGED`. This two-state lifecycle prevents repeated alerting from the same event.

AUTHENTICATION

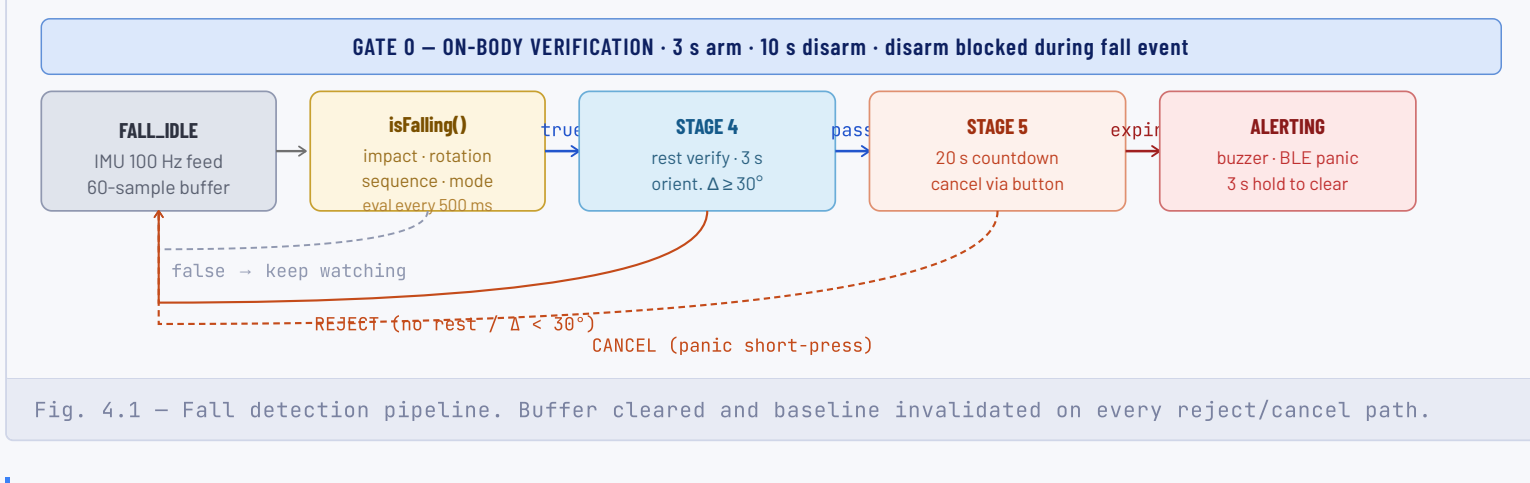
All POST endpoints require the header `Authorization: Bearer esp32_static_token_12345`. GET and acknowledge endpoints are presently unauthenticated, consistent with a closed local-network deployment where the server is not exposed to the public internet. The token is embedded in the phone app at build time.

04 LOCAL COMPUTING – FALL DETECTION

Fall detection is the most computationally intensive module and runs entirely on the ESP32 – no server round-trip is in the critical alert path. The algorithm is a **four-stage sequential state machine** evaluated every 500 ms, operating on a statically-allocated 60-sample ring buffer fed by the IMU at 100 Hz. Every stage imposes independent physical criteria; failure at any stage resets the pipeline to `FALL_IDLE` and clears the buffer.

GATE 0 – ON-BODY VERIFICATION (PRECONDITION)

Before any fall processing occurs, Gate 0 must confirm the device is worn on a live human body. It runs continuously at 200 ms cadence and arms the pipeline only when **either** the capacitive touch pad detects grip or the MAX30102 reports skin contact (`bodyStatus == 3`) with confidence $\geq 50\%$, sustained for a continuous **3-second window**. Disarming requires both sources to fail for **10 seconds** – an asymmetric hysteresis preventing momentary signal dropout from disarming the device mid-use. Disarming is blocked entirely while a fall event is being processed.



IMU RING BUFFER – ZERO-HEAP DESIGN

The original implementation used `std::queue<IMUData>`, backed by `std::deque`, which heap-allocates memory in chunks – fragmentation-prone on a 300 KB MCU. The replacement is a custom `IMURingBuffer<IMUData, 60>`: 720 bytes of BSS, `O(1)` push (evict-oldest when full), `O(1)` indexed read, `O(1)` `clear()`, and a `portMUX` spinlock for safe concurrent access between the 100 Hz ISR drainer and the 500 ms scheduler task. Direct indexing also eliminated the full-buffer copy previously required by `isFalling()`.

04 KINEMATIC PATTERN DETECTOR – ISFALLING()

`isFalling()` scans the full 60-sample ring buffer on every call, looking for three signals in a required temporal sequence. It is the single gate between `FALL_IDLE` and Stage 4.

THREE REQUIRED SIGNALS

SIGNAL 1 · Descent anchor

A temporal reference point marking the start of fall dynamics.

Two candidates are evaluated and the earlier one chosen:

Free-fall dip – accel magnitude below 0.73 g (< 3000 LSB). The crutch is briefly in free flight. Strongest physical indicator; used as anchor when present.

Peak rotation index – the sample of maximum gyro magnitude. Fallback for held-crutch falls, where the user's arm absorbs the descent and no weightlessness is observed. Always present when the gyro gate passes.

SIGNAL 2 + 3 · Impact + sequence

Impact spike – accelerometer magnitude must exceed a mode-dependent threshold. Represents ground contact after descent.

Correct sequence – the descent anchor index must precede the impact index, and the gap must be ≤ 50 samples (500 ms at 100 Hz). This prevents an isolated hard knock from being linked to an unrelated rotation elsewhere in the 600 ms window.

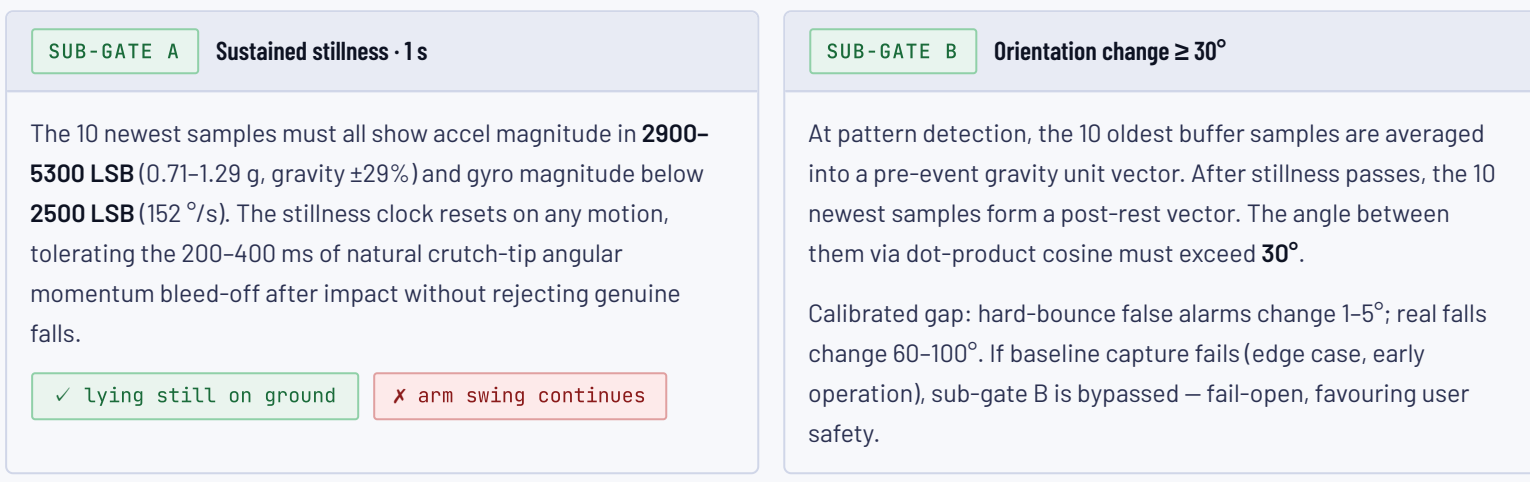
THREE-MODE THRESHOLD SELECTION

The single most impactful false-positive reduction came from conditioning thresholds on grip state. The touch sensor is read at each evaluation, selecting one of three paths:

Mode	Condition	Impact threshold	Gyro threshold	Rationale
HELD	Touch active	5700 LSB (1.39 g)	2480 LSB (150 °/s)	Arm absorbs impact; mild tip kinematics. Often no free-fall dip. Lenient thresholds needed to detect held falls.
STRICT-A	Unheld + free-fall	5700 LSB (1.39 g)	2480 LSB (150 °/s)	Free-fall dip alone is strong evidence. Modest impact + rotation sufficient.
STRICT-B	Unheld, no free-fall	8000 LSB (1.95 g)	5740 LSB (350 °/s)	No kinematic separator from hard knock. Recorded false alarm: 310 °/s – below gate. Real falls: 486+ °/s – above it.

STAGE 4 – POST-IMPACT REST VERIFICATION

Stage 4 opens a **3-second window** after pattern detection and requires two independent sub-gates to pass:



STAGE 5 – COUNTDOWN & ALERTING

On Stage 4 passing, a **20-second countdown** is displayed on the OLED. A fresh short press of the panic button (rising-edge detected, not level-held) cancels and resets the pipeline. After cancellation, a `WaitingForRelease` lockout prevents the still-held button from immediately re-arming the manual panic alarm. If the countdown expires, `sendPanic(isFall=true)` notifies the phone over BLE and the full three-phase buzzer alarm activates. Recovery requires a **3-second button hold**.

05 CONCLUSION

FORCE demonstrates that a production-quality elderly care monitoring system can be built on low-cost commercial embedded hardware, with all safety-critical computation executing locally on a microcontroller. The system successfully integrates continuous vital sign monitoring, fall detection, and caregiver alerting into a single wearable form factor – an ordinary forearm crutch – without requiring the user to carry or interact with additional devices beyond their existing walking aid.

TECHNICAL ACHIEVEMENTS

RELIABILITY

The multi-stage fall detection pipeline with three-mode threshold selection, orientation change verification, and Gate 0 on-body gating achieves significant false-positive suppression – rejecting hard-bounce knocks, deliberate crutch waving, and table-drop events that would trigger simpler threshold detectors.

ROBUSTNESS

Every module includes sensor-failure recovery: the bio hub and temperature sensor check readiness on every tick and attempt re-initialisation automatically. The BLE stack re-advertises on disconnect. The cooperative scheduler ensures no single task can block others. The ring buffer's portMUX spinlock guarantees ISR/task data integrity.

USER SAFETY

The 20-second self-cancel window respects user autonomy – avoiding unnecessary emergency dispatches – while the fail-open orientation bypass ensures a genuine fall is never silently discarded due to early-operation buffer edge cases. The escalating three-phase buzzer alerts bystanders even before the BLE notification arrives.

SYSTEM INTEGRATION

The full-stack architecture – embedded firmware, REST API, phone relay app, and caregiver dashboard – demonstrates end-to-end integration from raw sensor LSB values to a human-readable alert acknowledgement workflow. The panic event lifecycle (ACTIVE → ACKNOWLEDGED) provides a complete audit trail of every alert, with timestamps for both event creation and caregiver response.

LIMITATIONS AND FUTURE WORK

The current system operates on a local area network, limiting caregiver monitoring to the same Wi-Fi network. A production deployment would route the REST server through a cloud endpoint with TLS and proper authentication. The static Bearer token would be replaced with per-device certificate authentication. Threshold calibration was performed on a limited set of captured events; extended field testing across a broader demographic would allow statistical refinement of the STRICT-B gyro gate and the 30° orientation threshold.

Fall detection thresholds are currently fixed at compile time. An adaptive calibration pass – learning the device's resting orientation and typical gait signature from the first few minutes of use – would improve sensitivity for users with atypical gait or for crutches mounted at non-standard angles.

SUMMARY

FORCE is a functional, end-to-end elder care assistive device built within the constraints of a competition timeline. The firmware architecture, sensor integration, and fall detection algorithm represent production-quality engineering decisions grounded in physical measurement, empirical calibration, and explicit safety tradeoffs. All sensor housings, mounting hardware, and PCB enclosures were custom designed and 3D-printed by the team from Politehnica University of Timișoara.